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CS301 Quiz File

Jun 05, 2023 12:00 AM

1) A tree is an AVL tree if

Select the correct option

- A. Any one node fulfills the AVL condition
- B. At least half of the nodes fulfill the AVL condition
- C. Root node fulfills the AVL condition
- D. All the nodes fulfill the AVL condition

2) When a function calling itself is called as _____.

Select the correct option

- A. Inline
- B. Iteration
- C. Recursion
- D. Nested loop

3) What will be output of following code?

```
main() {
    int x = 15;
    int &y;
    y = x;
    y = 15;
    cout<<x;
}
```

Select the correct option

- A. 10
- B. 25
- C. **Compile time error**
- D. 15

4) Level-order traversal for Binary Search Tree can be implemented, _____.

Select the correct option

Through both recursive method call and non-recursive method call

Only through recursive method

Through max-heap

Only through non-recursive method

4) What will be the output of following code?

```
void explode(int& num) {
    num++;
}

main() {
    int power = 15;
    explode(power);
    cout<<power;
}
```

Select the correct option

- A. 14 B. 15
- C. 16
- D. 17

5) Which one is the correct function call for the following function of calculating cube? int
cube(int& num)

```
{  
.  
.  
.  
}
```

Select the correct option

- A. cube(num) B. cube(*num)
- C. cube(&num)

6) Consider an array int x[] = {14,15,4,9,7,18,3}. Which statement will be used to set first number of the list into the root node.

Select the correct option

- A. root->setInfo(&x[1])
- B. setInfo -> root(&x[0]);
- C. setInfo -> root(&x[1]);
- D. root->setInfo(&x[0]);

7) What will be output of the given C++ code?

```
char lhs[] = "Army"; char
```

```
rhs[] = "Army";
```

```
cout<<strcmp(lhs,rhs);
```

Select the correct option

- A. Syntax error
- B. Positive number
- C. 0
- D. Negative number

8) We allocate memory dynamically by using _____ operator.

Select the correct option

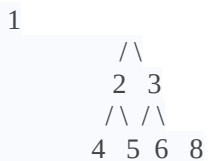
- A. increment
- B. new
- C. decrement
- D. this

9) Following is a keyword of C++ _____

Select the correct option

- A. delete
- B. eliminate
- C. del
- D. remove

10) What will be level order traversal of the given BST ?



Select the correct option

6 2 8 1 5 3 4

1 4 3 5 2 8 6

1 2 3 4 5 6

11) While implementing non-recursive traversal for Binary Search Tree, we need to implement _____.

Select the correct option

- A. Queue
- B. Stack
- C. Min heap
- D. Max heap

12) Suppose we have the following values to be inserted in constructing AVL tree,
20,23,25,10,12,13

Tell when first rotation will take place,

Select the correct option

- A. after inserting node 23
- B. after inserting node 25
- C. after inserting node 12
- D. after inserting node 10

13) Link List can be special case of,

Select the correct option

- A. Heap
- B. AVL Tree
- C. Binary Search Tree
- D. Parse Tree

14) In tree, the search operation is _____ as compared to the linked list.

Select the correct option

- A. very slow
- B. very fast
- C. equally time-taken
- D. not fast

15) The balance of a node in a binary tree is defined as the height of its _____ sub tree minus height of its right sub tree.

Select the correct option

- A. upper
- B. lower
- C. right

D. left

16) If the root of a tree is at level zero, its two children (subtrees) i.e. nodes will be at _____

Select the correct option

A. Level 1

B. Level 0

C. Level 3

D. Level 2

17) If a function has recursive call as the last statement, it is known as _____.

Select the correct option

A. Function recursion

B. Local Recursion

C. Last recursion

D. Tail recursion

18) Function signatures are also called :

Select the correct option

A. Function prototype

B. Function definition

C. Function overriding

D. Function overloading

19) if p is pointing to a node in an existing tree then info(p) means:

Select the correct option

A. Returns false if it is not a parent node

B. Returns the contents of node p

C. Returns the pointer of node p

D. Returns true if it is a parent node

